Whether Small Transformers Copy a Feature from Context Depends on the Noun's Training History

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Abstract

Small transformers trained on text in which nearby sentences describe the same kind learn to copy a feature from one sentence into the next. Probes of this use nonce nouns, which are unfamiliar and have untrained representations. In two registered experiments with 150 models (3.4M to 25.6M parameters) trained on an artificial language, we controlled how often and how 40 nonce nouns appeared. With stable shapes, unseen nouns copied a shape from the preceding sentence, but mostly because of their untrained input rows. Familiar nouns whose sentences came one per window stopped copying within about 100 exposures, whatever their shapes. Nouns whose sentences came in runs that shared a shape, with the shape changing between runs, kept copying: 3.4 nats more than the same history presented singly (registered), and 2.9 nats more at 1,000 exposures than runs with a fixed shape (exploratory). Copying did not depend on whether the context sentence was about the same noun. In pretrained Pythia models, replacing nouns' input rows had no consistent effect on copying colour. In the small models, a familiar noun kept triggering copying only if, in training, its own earlier sentences had been the sole source of its current shape.

1 Introduction

A language model can take a word's properties from the text around it or from what it has stored about the word. Which it does depends on its training data. Classes that occur in bursts, or whose meaning changes, favour learning in context, and common, stable classes favour learning stored in the weights, both across training distributions and between rare and common classes within one model (Chan et al., 2022, 2025). In pretrained models, mechanisms that copy from the context compete with recall from memory (Yu et al., 2023; Jin et al., 2024; Ortu et al., 2024). We ask how this plays out for single words: as a model sees a noun more often, when does it stop copying the noun's features from the context, and what about the noun's training decides this?

Probes with nonce words cannot answer this alone, because a nonce word is new in two ways: it has nothing stored about it, and its representation is untrained or unusual (Schick and Schütze, 2020; Land and Bartolo, 2024). Conclusions drawn with nonce words can depend on how the new word is represented (Kim et al., 2022; Cacioli, 2026), and we found this in our own earlier experiments (Section 3): small transformers carried a shape from one mention of an unseen noun to the next, but most of that carry vanished when the noun was given an input embedding typical of trained nouns.

We therefore controlled, for each of 40 nonce nouns, how many times it appeared in training, with what shapes, and how its sentences were arranged, and registered the hypotheses before training (Appendices G and H). The second experiment tests two confounds found in the first. In models trained with stable shapes, we find:

1. **Unseen nouns copy because their rows are unusual.** A typical input row removed most of their carry; so did moving the row to the trained nouns' mean (Experiment 1; row analysis).

2. **Familiar nouns presented singly stop copying.** By 100 exposures carry was under 1 nat, whether a noun had been seen with a fixed shape, a changing shape or no shape (Experiment 1).

3. **Nouns whose earlier sentences were the only source of their shape keep copying.** With sentences in runs that shared a shape, and the shape changing between runs, carry stayed high at 1,000 exposures. Runs with a fixed shape, or no shape, did not keep it, and with frame and expected word class matched a stored shape did not lower carry (Experiment 2).

4. **The copying is not specific to the noun.** Carry was the same when the context sentence was about another noun with the same history.

In pretrained Pythia models, replacing real nouns' input rows had no consistent effect on how much colour they copied (Experiment 3). Our claims are about transformers, not children.

2 Related work

In-context and in-weights learning. In-context learning emerges when items occur in bursts, more when their meanings change, and trades off against learning in the weights (Chan et al., 2022); it depends on task diversity (Raventós et al., 2023), favours rare classes and can fade as stored solutions take over (Chan et al., 2025; Singh et al., 2023), and its onset has been traced to induction heads (Olsson et al., 2022; Reddy, 2024). Resetting embeddings during training keeps in-context use alive for rare and unseen tokens (Anand et al., 2025), and in synthetic biographies the most frequent entities shift from context to stored knowledge (Kim et al., 2026). We vary, for single words within one model, how often a word occurs, what is stored about it and whether its own contexts are informative.

Context against memory. Models answer from memory more for frequent entities (Mallen et al., 2023; Kandpal et al., 2023; Razeghi et al., 2022), resist contexts that contradict what they know (Longpre et al., 2021; Xie et al., 2024; Du et al., 2024), and copying and recall compete inside the network (Yu et al., 2023; Jin et al., 2024; Ortu et al., 2024). Here frequency is observed, not manipulated.

Nonce words, rare words and new tokens. Nonce words are the usual test of generalisation to a new item (Thrush et al., 2020; Misra et al., 2023). Rare and new words have poor representations (Schick and Schütze, 2020), under-trained tokens behave abnormally (Land and Bartolo, 2024), lexical exposure can inflate apparent generalisation (Kim et al., 2022), and controlling how often an item occurs shows how frequency shapes what transformers learn (Wei et al., 2021). We control exposure down to the window in which each occurrence is read.

3 Setting and earlier results

Language and models. Each corpus describes 32 kinds of object, each named by a nonce noun, with 14 exemplars per kind. Each exemplar has a shape, a colour and a texture, each one of ten nonce

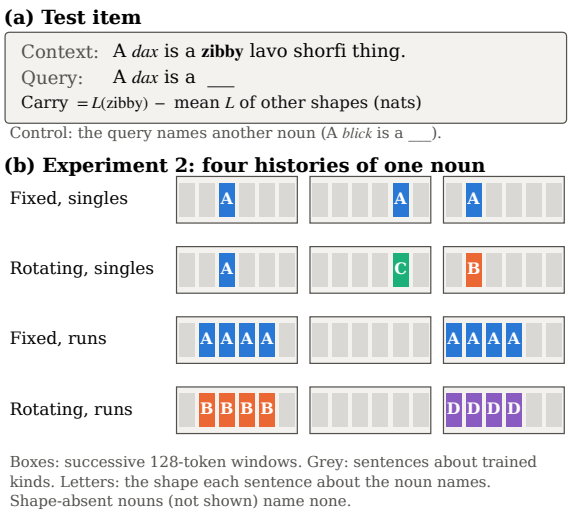


Figure 1: (a) A test item. (b) Four of the histories of Experiment 2, drawn as the windows in which a noun's sentences are read.

words, and is described in one of several frames, such as *A wug is a zibby lavo shorfi thing* (shape, colour, texture; Appendix E). In **Regular** corpora each kind keeps one shape. In **Reassigned** corpora a kind gets a new shape in every pass through the corpus and keeps it within the pass, so only the context can tell a trained noun's current shape. In **grouped** order, sentences about a kind come in runs of four; in **random** order they are shuffled. Text is read once, in non-overlapping 128-token windows. Models are pre-LayerNorm decoder-only transformers with learned position embeddings, of 3.4M, 11.0M and 25.6M parameters (4 to 8 layers; Appendix C). Every artificial word is one token.

Carry. A test item gives one context sentence about a noun, naming shape s, then the query *A noun is a*, and reads the log probability L of each of the ten shapes at the next position (Figure 1a). **Carry** is $L(s)$ minus the mean of L over the other shapes, excluding the noun's own shape s^* (a placebo shape for nouns without one); s is never s^*. Each noun gets 36 items, each of the nine shapes other than s^* four times. A carry of 3.4 nats means the context's shape is about 30 times as probable as an average other shape. In a **control**, the query names another noun with the same history, and **noun-specific carry** is carry minus control carry. **Knowledge** is $L(s^*)$ minus the mean over the other nine shapes after an unrelated sentence.

Earlier results. In registered experiments with 120 models (Appendix A), grouped training raised a related index of carry for unseen nouns over ran-

dom order by 11.08 nats [9.57, 12.59]. But the unseen nouns' input embeddings had never been trained, and when the same nouns were given an embedding typical of trained nouns (their mean plus a random offset of typical length; compare initialising new embeddings at the mean, Hewitt, 2021), the effect shrank to 3.00 nats [1.79, 4.21] (exploratory; Appendix F). So these probes measured arrangement together with the test noun's representation. Below, input and output embeddings are untied by design.

4 Experiment 1: exposure and history

Design. We added 40 nonce nouns to three base corpora (Regular grouped, Regular random and Reassigned grouped). In every model they fill 13 cells: three histories, each at 10, 100, 1,000 and 10,000 exposures, three nouns per cell, plus four unseen nouns. A *fixed-shape* noun always appears with its own shape s^*; an *uninformative* noun with a new shape every sentence, drawn from the corpus's shapes; a *shape-absent* noun only in a frame that names no shape (*The noun is colour and texture*, which trained kinds also use). A trained kind appears in about 60,000 sentences. Each familiarity sentence lies wholly inside one window, at most one per window, between runs in grouped corpora; a noun's exposures are evenly spaced from 5% to 95% of training. A checker verified every exposure from the tokens, and the corpora matched checksums recorded before training. We trained 90 models (three sizes, ten seeds, three base corpora). The experiment was registered before any of its corpora was built.

Analysis. The seed is the unit of replication: we average cell means over the three sizes within each seed and test over ten seeds, with Holm (1979) correction within each experiment. A hypothesis is supported only if its direction also holds at two of the three sizes. Hypotheses are numbered as in the registrations.

Results. Fixed-shape nouns learned their shape (knowledge at 1,000 exposures 8.3 to 9.5 nats in the three corpora; M1 passed). In grouped Regular models, unseen nouns carried 4.53 nats [3.54, 5.52], but only −0.33 with a typical row. Familiar nouns lost carry for every history (Figure 2, left): 3.6 to 4.5 nats at 10 exposures, 0.16 to 0.86 at 100, and within 0.4 nats of zero from 1,000 on. Grouped models carried more than random ones for unseen nouns (4.68 [3.71, 5.65]), and in random order carry was near zero for every noun.

We predicted that carry would fall faster for fixed-shape than for shape-absent nouns (H30). It did not (slope difference 0.16 nats per decade [0.01, 0.30], the wrong way). Fixed-shape nouns' stored preference for s^* rose by 2.97 nats per decade (H32), yet they carried within 0.4 nats of uninformative nouns at every exposure (exploratory). Familiar shape-absent nouns at 1,000 and 10,000 exposures carried 0.33 nats [0.13, 0.52] more than unseen nouns with a typical row; the difference was reliable, and the two were also equivalent within the registered 1-nat bound (H31). In Reassigned models (exploratory), familiar nouns kept more carry (shape-absent: about 2 nats at 1,000 and 10,000 exposures; Figure 2, middle). Carry was not specific to the noun: in every Regular cell, noun-specific carry was within 0.14 nats of zero (Table N6).

A registered head search did not localise carry. We ablated the heads with the highest product of copying and prefix-matching scores, the signature of induction heads (Olsson et al., 2022); no head showed clear prefix matching, so these were copying heads, nearly all in the first layer. Ablation removed a mean proportion of 0.22 [0.07, 0.36] of unseen carry, against a registered 0.5, and not reliably more than random heads (H33 not supported). In Reassigned models it cut trained kinds' carry by 7.41 nats (H34 supported), but random heads cut it by 6.91, so the cut is not specific (Appendix G).

Two confounds. A blind review of this experiment (Appendix H) noted two differences between familiar nouns and trained kinds other than familiarity. A familiar noun never recurred within a window, while trained kinds came in runs, where an earlier sentence predicts the next. And shape-absent nouns never appeared in the test frame.

5 Experiment 2: recurrence and useful context

Design. The 40 nonce nouns were reassigned to 12 cells, three nouns each, plus four unseen nouns: three histories × two presentations × 100 or 1,000 exposures (Table 1; Figure 1b). Presented *singly*, a noun has one sentence per window. Presented in *runs*, it has four consecutive sentences in one window, so runs and singles are matched in sentences but runs give a noun a quarter as many windows. Shape-absent nouns now use a frame that begins like the test frame but names no shape (*A noun is a colour texture thing*), which trained kinds also

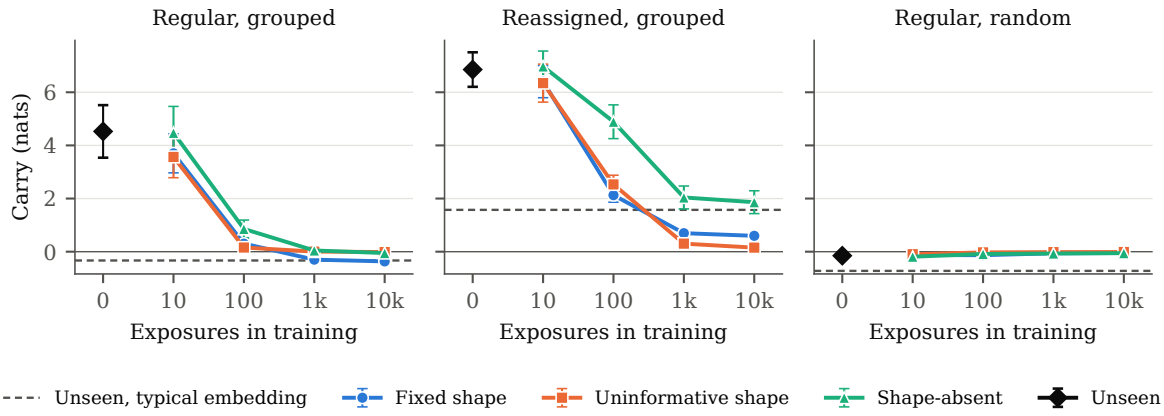


Figure 2: Experiment 1: carry of the context's shape by exposure and noun history (nats; means over ten seeds of size-averaged cell means, 95% intervals). Diamond: unseen nouns. Dashed line: unseen nouns with a typical input row. Uninformative nouns are Experiment 2's rotating nouns presented singly.

History	Sentences about the noun in training
Fixed	always its own shape s^*
Shape-absent	*A noun is a colour texture thing*: no shape
Rotating	a shape drawn from the corpus's shapes: a new one every sentence when presented singly; one per run, changing between runs, when presented in runs

Table 1: Experiment 2. Each history is presented singly or in runs of four, at 100 or 1,000 exposures. Only in rotating runs is the noun's own earlier sentence the sole source of its current shape.

use in one sentence in fifteen. Everything else is as in Experiment 1. We trained 60 models on Regular grouped (confirmatory) and Reassigned grouped (exploratory) corpora, three sizes and ten seeds. The experiment was registered after its corpora were built and checksummed and before any of them was used for training.

Hypotheses. We registered: runs and singles differ for shape-absent nouns (H40) and fixed nouns (H41), each two-sided with equivalence bounds of ± 0.3 nats; rotating runs carry more than rotating singles (H42); fixed singles carry less than shape-absent singles (H43); shape-absent singles differ from unseen nouns with a typical row (H44, two-sided, ± 0.3). A check (K1) required fixed nouns' knowledge to be equivalent within ± 1 nat between runs and singles; otherwise H41 could reflect a change in what is stored rather than recurrence. The 100- and 1,000-exposure cells are pooled. Splits by exposure, matched contrasts and the control are exploratory.

Runs that made the context the only source kept carry. Unseen nouns carried 6.35 nats [5.59, 7.12] (M2 passed). Rotating nouns in runs carried 3.39 nats [2.69, 4.09] more than rotating nouns presented singly (H42 supported; Figure 3): 3.43 at 100 exposures and 3.35 at 1,000. Runs and singles differ in the number of windows, but rotating runs also carried more than fixed runs, which are matched in frame, presentation and windows: 1.73 nats [1.25, 2.20] pooled, and 2.86 [1.99, 3.74] at 1,000 exposures (Table N5). At 1,000 exposures rotating runs still carried 3.85 nats, against 0.50 to 1.03 for every other familiar cell (Table N2).

Recurrence alone helped less. Shape-absent nouns in runs carried 0.80 nats [0.47, 1.13] more than in singles (H40, "above"): 1.55 at 100 exposures and 0.05 [−0.30, 0.40] at 1,000. Fixed nouns in runs carried 1.69 [1.29, 2.08] more (H41, "above"), but K1 failed: at 100 exposures runs weakened the stored shape (knowledge 3.0 against 5.4 nats; Table N3). By the registered rule we do not read H41 as a recurrence effect. At 1,000 exposures, where knowledge was equivalent, the difference was 0.47 [0.27, 0.66].

A stored shape did not lower carry once expectations were matched. Fixed singles carried 1.02 nats [0.76, 1.27] less than shape-absent singles (H43 supported), and shape-absent singles carried 1.42 [1.12, 1.73] more than unseen nouns with a typical row (H44, "above", not equivalent; 2.48 at 100 exposures, 0.37 [0.15, 0.59] at 1,000). But shape-absent nouns, trained in a frame where a colour follows *is a*, expected a colour rather than a shape at the test position, whereas every other familiar noun expected a shape (Table N3). Fixed and rotating singles share frame, presentation and

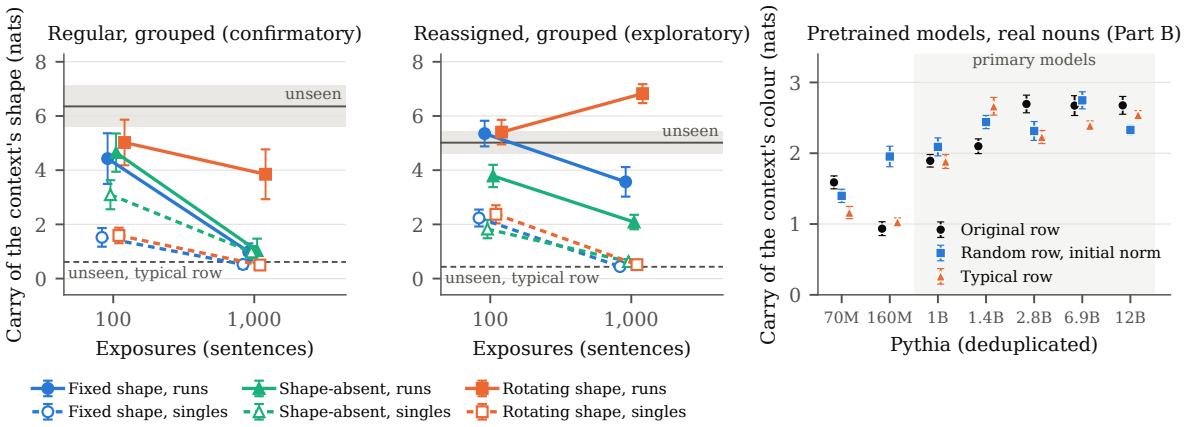


Figure 3: Left and middle, Experiment 2: carry of the context's shape by history, presentation and exposure (nats; means over ten seeds of size-averaged cell means, 95% intervals). Solid line and band: unseen nouns; dashed line: unseen nouns with a typical input row. Right, Experiment 3: carry of the context's colour for real nouns in Pythia models with their original input rows or replaced rows (a different measure, in nats; means over 44 nouns, 95% intervals).

expected word class, and differ only in whether one shape is stored; their carry was the same (−0.02 [−0.16, 0.11]; Table N5). H43 thus reflects what the noun predicts at the test position, not a stored shape.

Not specific to the noun. In Regular models, noun-specific carry was at most 0.07 nats in every cell, including rotating runs (Table N6): the model copied the context's shape after the query noun whichever noun the context was about.

Reassigned models (exploratory). Run effects were larger (H40 1.71, H42 4.67), rotating runs at 1,000 exposures carried more than unseen nouns (6.82 against 5.02), and fixed and shape-absent runs also kept carry (3.57 and 2.09), though for fixed nouns runs again weakened the stored shape. Here part of carry was noun-specific (0.48 for unseen nouns, 1.45 for rotating runs at 1,000). Overall carry was higher than in Experiment 1 (unseen 6.35 against 4.53 nats), although the base corpora differ only in the frame without a shape; the other nonce nouns' histories may affect carry, so we compare conditions only within an experiment.

6 Row analysis

A noun enters the model only through its input row. In Experiment 1's grouped models we replaced only input rows and scored carry again; unchanged rows reproduced the original scores. In Regular models, scaling an unseen noun's row to the trained kinds' mean norm, keeping its direction, lowered carry from 4.53 to 4.00 nats (difference 0.52 [0.20, 0.85]; H49 supported), almost entirely in the 3.4M models. A random direction at that norm gave the same carry (4.09), and the trained kinds' mean row removed it (–0.44). Shrinking a familiar shape-absent noun's row (10,000 exposures) to the unseen nouns' norm raised its carry only from –0.05 to 0.53. So the norm explains little; carry needs a row away from where trained rows lie. In Reassigned models (exploratory) rescaling raised unseen carry (by 0.27), and shrinking a familiar row raised its carry from 1.86 to 4.16 (Table N8).

7 Experiment 3: pretrained models

Design. This experiment asks only whether the row result transfers: do pretrained models copy more for nouns given an untrained-like row, or less for nouns with a typical colour? It does not manipulate history. We used Pythia deduplicated models (Biderman et al., 2023) (final checkpoints; 1B to 12B primary, 70M and 160M descriptive, 410M pilot). Forty-four one-token nouns formed 22 pairs of a colour-typical noun (e.g., *sky*, *leaf*, *bone*) and a colour-variable noun (e.g., *truck*, *hat*, *dog*), matched on count/mass type and on frequency in the Pile (Gao et al., 2020), counted with infini-gram (Liu et al., 2024). Text-trained models' colour knowledge follows how colours are reported in text (Paik et al., 2021), so we also checked that typical colours were stored. Two passes of an AI coder with a fixed prompt rated colour typicality before any model was run (weighted κ =

0.96). Each noun was read in three frames such as *The noun was colour. The noun was*, with the 11 basic colour terms that are one token; carry is the mean over frames and ordered colour pairs of $\log p(c_1 \mid \text{context } c_1) - \log p(c_1 \mid \text{context } c_2)$, renormalised over colours. We replaced only each noun's input row, with a random direction at the norm of an initial row (*atypical_init*) or a typical row. We registered that carry would be higher with atypical_init rows than with original (H45) or typical rows (H46), lower for colour-typical nouns (H47), and that the H45 increase would differ between groups (H48); each needed Holm-corrected significance, within each model, in four of the five.

Results. Colour-typical nouns did prefer their colour with no context (by 1.3 to 1.7 nats, for 82 to 95% of nouns; Table N11). With their own rows, nouns carried 1.89 to 2.70 nats of colour. No hypothesis was supported. Atypical_init rows changed carry by −0.38 to +0.34 nats against original rows, Holm-significant and positive in two models (1B, 1.4B) and negative in two others; against typical rows the pattern differed (positive and significant in 1B and 6.9B, negative in 1.4B and 12B). Colour-typical and colour-variable nouns did not differ reliably (−0.16 to 0.11), and H48 was inconclusive (Table N9). The changes were mostly not equivalent to zero within ± 0.3 nats either, so the result is inconsistent, not null. Changing only the norm of a replaced row changed carry by under 0.01 nats on average in every primary model. A mixed model over the five models gave no effect of either row or of group.

8 Discussion

In our small transformers trained with stable shapes, whether a familiar noun triggered copying of the preceding sentence's shape depended on its training history. Nouns presented singly stopped within about 100 exposures in Experiment 1, whatever they had been seen with. Nouns presented in runs that shared a shape, with the shape changing between runs, still carried 3.85 nats at 1,000 exposures; runs with a fixed shape or no shape carried about 1 nat. Our reading is that copying persisted where the noun's own earlier sentences were the only reliable source of its current shape. These are the conditions, bursty items whose meaning changes, that favour in-context learning for classes (Chan et al., 2022); here they act on single nouns within one model. Experiment 1's apparent conclusion, that familiarity switches copying off, holds only for nouns whose contexts never made copying useful.

Two qualifications follow. First, the copying was not bound to the noun: in Regular models a noun's history decided whether the model copied the last shape, not whether it tracked that noun's shape, and only Reassigned training produced a noun-specific part. Second, the noun's history interacts with the rest of the corpus. In Reassigned models, where trained kinds teach that a stored shape can be wrong, runs kept carry even with a fixed or no shape, and overall carry differed between our two experiments.

Stored knowledge itself mattered less than we expected: with frame and expected word class matched, a stored shape did not lower carry relative to a shape that changed every sentence. Where it may matter is in runs: fixed runs, whose earlier sentences predicted a shape that memory also supplied, lost carry, and rotating runs did not.

Unseen nouns are not a neutral baseline. Their carry depended on rows unlike those of trained nouns, which fits the finding that untrained embeddings preserve in-context use (Anand et al., 2025). Effects measured with nonce tokens should be checked with representations that look familiar. In Pythia, replacing a real noun's row did not change colour copying consistently, and the row's norm made almost no difference, so the untrained-row effect did not transfer to these models and this probe. Whether pretrained models' copying depends on a noun's burstiness in pretraining remains to be tested.

Limitations

The language is artificial, with 32 kinds, three dimensions and a few frames; the small models are trained with one recipe, and effects were largest in the 3.4M models (Table N7). We tested carry only in the frame and dimension that rotating-run nouns were trained on, one sentence back; whether it transfers to other frames, to colour or to longer gaps is untested. Runs and singles differ in the number of windows; the matched contrast with fixed runs addresses this for the main result but not for H40. The other nonce nouns may affect a noun's carry, and we did not train models with a single history. We have no mechanism: the head search found no specific heads. Power for a 1-nat difference in Experiment 1 was planned at 0.57.

The pretrained test used one model family, 22 noun pairs, three frames and colour only, did not separate contexts that agree or conflict with a noun's typical colour, and its colour codes came from two passes of an AI coder, not human raters. Earlier experiments were registered in stages, some after related results were known (Appendix A). Experiments 1 and 2 were registered before training, and every registered test is reported.

Ethics statement

All training data are artificial; the pretrained models are public. No data were collected from people. Code, corpora, results and all trained models will be released; anonymised copies are provided as supplementary material.

Use of AI assistants

An AI assistant (Claude, Anthropic) was used to help design analyses, write code, review code and design, draft and edit text and check citations; to carry out the blind review that motivated Experiment 2 (it was given only the write-up of Experiment 1); and, in two independent passes, to code the nouns of Experiment 3. The authors are responsible for all content, including the accuracy of every citation.

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A Registration record

In the registrations, Study 1 is the earlier training experiments (Section 3; Appendices C and E), Study 2 an analysis of natural corpora and Study 3 an earlier test of pretrained models. Studies 2 and 3 do not bear on the claims of this paper; their methods and full results, and further results of Study 1, are in the long version in the supplementary material. Table A4 reports every registered hypothesis of all three studies.

All registrations are in one public project on the Open Science Framework (link withheld for review; an anonymised view-only link is in the supplementary material), and each states what was known when it was written. Table A1 lists them, Table A2 gives the label each registration uses for each row of Table A4, and Table A4 gives every registered hypothesis and its result.

Table A1: Registrations. Each was written before the models it concerns were trained or scored, or before the transcripts were read.

Registration	Written	Rows of Table A4	Known when it was written
Original	2 Mar 2026	Corpora, models and controls (Appendix D)	Nothing trained
Amendment 1	7 Oct 2026	1, 2, 3, 6, 16	Random-order models trained and scored on an earlier test, with the old loader; no grouped model trained
Amendment 2	7 Oct 2026	7 (texture added as a co-primary test)	No grouped model trained
Amendment 3	8 Oct 2026	23, 24, 25	Study 1 results for rows 1 to 3, 6, 7 and 16 (old loader); no pretrained model scored
Amendment 4	8 Oct 2026	20, 21	Studies 1 and 3 results; no transcript read
Amendment 5	9 Oct 2026	4, 5, 8, 9, 17, 18, 19, 22	All earlier results (Study 1 with the old loader); four reviews of a draft; five exploratory checks, including the confound of the noun with another
Amendment 6	11 Oct 2026	11, 12, 13	Amendment 5 results (old loader); a prediction for the new models fitted to them
Amendment 7	12 Oct 2026	10, 14, 15	All results, with corrected scores
Amendment 8	15 Oct 2026	26 to 32	All results, with corrected scores; three rounds of reviews, which noted that fixed-file models memorised their files and that the novel-noun test was uninformative; no fresh-text model trained
Amendment 9	23 Oct 2026	Familiarity study (Appendix G)	All results above; no familiarity corpus built
Amendment 10	24 Oct 2026	Recurrence, frames, rows and pretrained models (Appendix H)	Amendment 9 results and a blind review of them; no Amendment 10 corpus used, model trained or pretrained model run on its items

Table A2: Rows of Table A4 and the labels used in the registrations.

Row	Label	Row	Label	Row	Label	Row	Label
1	H4b	9	H13b	17	H14a	25	H8
2	H1b	10	H22	18	H14b	26	H23
3	H2b	11	H17	19	H15	27	H24
4	H11	12	H18	20	H9	28	H25
5	H12	13	H19	21	H10	29	H26
6	H5b colour	14	H20	22	H16	30	H27
7	H5b texture	15	H21	23	H6	31	H28
8	H13a	16	H3b	24	H7	32	H29

The loading correction. The model code ties the input embedding to the output layer. On the graphics

card used for training (through DirectML) the tie was lost, so every Study 1 model was trained and saved with two separate matrices. The first version of the scoring code rebuilt each model with the tie in place before loading its weights, so the output matrix overwrote the input embedding. We found this on 12 October 2026, fixed the loader, and rescored all 315 Study 1 models then trained, 461 intermediate checkpoints of the random and grouped Regular models and the trained-noun items, with the same items. A check on eight randomly chosen models confirmed the fix: mean loss on each model's own training corpus was 0.07 to 0.16 nats per token with the corrected loader, close to the last logged training loss (0.08 to 0.23; dropout is off at test), and 0.58 to 1.97 with the old one. Studies 2 and 3 were not affected. The old scores are kept in the repository.

Table A3: Study 1 registered tests before and after the correction (shape-over-colour index, nats). Rows 10, 14 and 15 were registered and scored after the correction.

Row	Old loader	Corrected	Verdict before → after
1	1.36 vs 0.18	6.49 vs 0.47	Supported → supported
2	0.18	0.47	Not supported → not supported
3	0.18 vs 0.09	0.47 vs −0.25	Not supported → supported
4	difference 1.45	difference 6.93	Supported → supported
5	difference 1.17	difference 2.76	Supported → supported
6	1.36 vs 0.80	6.49 vs 2.12	Supported → supported
7	1.10 vs −0.04	5.36 vs 0.46	Supported → supported
8	slope 0.29	slope 1.63	Supported → supported
9	0.24 vs 0.18	1.43 vs 0.47	Not supported → supported
11	0.21 vs 0.18	0.50 vs 0.47	Not supported → not supported
12	0.29 vs 0.24	0.80 vs 1.43	Not supported → not supported (reversed)
13	0.21 vs −0.08	0.50 vs 0.10	Supported → supported
16	difference 0.0007	difference 0.005	Equivalent to zero → equivalent to zero

Table A4
All registered hypotheses and results; registered secondary analyses are reported in the text. Rows 1 to 16 concern fixed-file models and rows 26 to 32 fresh-text models. Seed-level tests average the three sizes within each of five seeds (t-based 95% intervals, two-sided; tests one-sided unless marked, 4 degrees of freedom), so a one-sided test can be below .05 while the two-sided interval includes zero (row 13). The registered p is the test named in the registration, one-sided unless marked, Holm-corrected within its registration; for rows 26 to 32 it is the seed-level test, corrected within rows 26 to 29 and within rows 30 to 32. Corpus intervals resample sources and then documents; Pythia and BabyLM intervals resample nouns.

#	Prediction	Result (nats)	Seed-level difference [95% CI]	Seed-level p	Registered p	Verdict
	1. Does grouping teach the fixed feature?					
1	Grouped order gives a higher index than random order	6.49 vs 0.47	6.02 [5.24, 6.79]	$< .001$	$< .001$	Supported
2	Random order gives an index above zero	0.47	0.47 [−0.61, 1.56]	.15	.061	Not supported
3	Random order: Regular above Mixed-slot	0.47 vs −0.25	0.73 [−0.69, 2.15]	.11	.032	Supported by the registered test, not at the seed level
4	Grouped: Regular above No stable feature	6.49 vs −0.44	6.93 [5.47, 8.40]	$< .001$	$< .001$	Supported
5	Grouped: Colour stable below No stable feature	−3.20 vs −0.44	2.76 [1.99, 3.52]	$< .001$	$< .001$	Supported
6	Grouped: Regular above Mixed-slot, shape over colour	6.49 vs 2.12	4.37 [1.98, 6.76]	.004	$< .001$	Supported
7	Grouped: Regular above Mixed-slot, shape over texture	5.36 vs 0.46	4.89 [2.69, 7.10]	.002	$< .001$	Supported
8	The index rises with log B across six orders	Slope 1.63 (SE 0.10)	1.63 [1.46, 1.80]	$< .001$	$< .001$	Supported
9	B = 3.6 above random order	1.43 vs 0.47	0.96 [0.63, 1.28]	$< .001$	$< .001$	Supported
10	The same, Objects corpora, different-object runs	1.18 vs 0.22	0.96 [0.37, 1.54]	.005	$< .001$	Supported
26	Fresh text: grouped order gives a higher index than random order	9.52 vs −1.56	11.08 [9.57, 12.59]	$< .001$	$< .001$	Supported
27	Fresh text: B = 3.5 above random order	1.28 vs −1.56	2.84 [1.41, 4.27]	.003	.005	Supported
28	Fresh text, grouped: Regular above No stable feature	9.52 vs −3.98	13.50 [11.48, 15.52]	$< .001$	$< .001$	Supported
	2. What kind of grouping helps?					
11	512-token window above 128, random order	0.50 vs 0.47	0.03 [−0.44, 0.50]	.44	.87	Not supported
12	512-token window above 128, B = 3.6	0.80 vs 1.43	−0.63 [−0.90, −0.37]	.003 (two-sided)	1.00	Not supported; reliably reversed
13	512-token window: Regular above No stable feature	0.50 vs 0.10	0.40 [−0.04, 0.85]	.032	.045	Supported; weak
14	Grouped: different-object above same-object runs	5.65 vs 2.99	2.66 [1.57, 3.75]	.001	$< .001$	Supported
15	B = 3.6: different-object above same-object runs	1.18 vs 0.45	0.73 [0.15, 1.32]	.013	$< .001$	Supported
29	Fresh text, grouped: different-object above same-object runs	7.60 vs 5.87	1.72 [0.54, 2.90]	.008	.008	Supported
	3. Is the expectation tied to the noun?					
16	Fixed files, random order: index higher after the same noun (novel nouns)	Difference 0.005; equivalent to zero ($p = .001$), but the novel nouns were untrained (Section 3)			.46	Not supported
30	Shape reassigned, grouped: noun-specific index above zero, novel nouns	0.37	0.37 [0.24, 0.49]	$< .001$	.001	Supported

#	Prediction	Result (nats)	Seed-level difference [95% CI]	Seed-level p	Registered p	Verdict
31	The same, above grouped Regular (fresh text)	0.37 vs 0.12	0.25 [0.10, 0.39]	.005	.005	Supported
32	Shape reassigned, grouped: noun-specific index above zero, nouns from training	11.24	11.24 [10.44, 12.03]	$< .001$	$< .001$	Supported
17	Pythia: noun-specific index rises with size (wording matched)	$\rho = -.42$			1.00	Not supported
18	Pythia 410M to 12B: noun-specific index above zero (wording matched)	−0.15 (SE 0.12)			1.00	Not supported
19	Pythia: 410M above 160M across ten training seeds 4. What do natural input and pretrained models show?	0.15 vs −0.15			.012	Supported
20	Speech to children: B above 1	3.57 [3.13, 4.30]			$< .001$	Supported
21	Speech to children above written text	3.57 vs 1.43; difference 2.14 [1.55, 2.91]			$< .001$	Supported
22	Speech to children differs from adult conversation (two-sided)	3.57 vs 2.33; difference 1.24 [0.51, 2.03]			$< .001$	Supported; higher
23	Pythia: index rises with size	$\rho = .92$			.003	Supported
24	Pythia: index rises over training	Mean within-size $\rho = -.34$			1.00	Not supported
25	BabyLM baselines (10M, 100M words): index above zero	0.47 [−1.05, 1.93]; 0.37 [−0.04, 0.73]			.52; .12	Not supported

Fresh text and shape reassignment. Rows 26 to 32 were registered on 15 October 2026, after reviews of the manuscript had noted that fixed-file models memorised their files and that the test with novel nouns was uninformative, and before any fresh-text model was trained. The corpora were built, and their checksums registered, before training. Amendment 8 names the conditions by their folder names (for example reassigned_grouped).

B Deviations from the registrations

1. Two registered analyses are not reported here: the Bayesian reference model of Amendment 1, which is in the repository, and the prediction of Amendment 6 for the 512-token models, which had been fitted to scores later corrected (Appendix A).

2. Added after registration: the seed-level tests for rows 1 to 3 and 6 to 8, the seed-level intervals, and intervals for the differences between natural corpora (from the registered two-stage bootstrap). Effective clustering was computed before it was registered as the scale for a figure of the long version and is reported there as exploratory.

3. The two models of about 70 billion parameters were scored with Apple's MLX library, because PyTorch crashed the machine; on smaller models the two differed by 0.003 to 0.010 nats on average.

4. In the hand coding of Study 2, a rule for labels not used for a physical object was added during coding and applied by both coders, and one pair of the reliability subset was discussed with an AI assistant and left out of the agreement check.

Smaller technical changes (software versions, download addresses, a stale file in one public model repository) are listed in the repository.

C Model and training settings

Table C1: AdamW (betas 0.9 and 0.95, weight decay 0.1), linear warm-up then cosine decay, dropout 0.1, untied input and output embeddings. Fixed-file models read windows starting at random positions in the corpus; fresh-text models read non-overlapping windows, each at most once, in random order (the 3.4M and 25.6M models used about 60% of the windows available, the 11.0M models 94 to 96%). Byte-pair tokeniser with 351 entries for the Regular corpora and 314 to 339 for the other corpus families; every artificial word is one token. Seeds 42, 123, 456, 789 and 1001 each set the corpus, the initial weights and the batch order. These five seeds are those of the earlier experiments; Experiments 1 and 2 used ten (adding 2718, 3141, 1618, 1414 and 1732), untied embeddings by design, a tokeniser of 329 entries and weights saved every 1,000 steps. All models are pre-LayerNorm decoders with learned absolute position embeddings, GELU feed-forward layers and a final LayerNorm. Checkpoints were saved every 500 steps. Training ran on one Windows PC through DirectML.

Setting	3.4M	11.0M	25.6M
Layers, heads, width	4, 4, 256	6, 8, 384	8, 8, 512
Feed-forward width	1,024	1,536	2,048
Steps	5,000	8,000	10,000
Batch (128-token windows)	32	32	16
Batch (512-token windows)	8	8	4
Peak learning rate	3×10^{-4}	2×10^{-4}	1×10^{-4}
Warm-up steps	200	400	500

Table C2: Memorisation. Mean next-token loss (nats per token) on 300 windows from each model's own training text and from a new arrangement of the same sentences (100 further passes, 27 for the Objects corpora, never used in training); 15 models per cell. At the seed level, the gap between the two losses was larger for fixed-file than for fresh-text models by 0.57 to 1.16 nats, with every interval excluding zero.

Corpus	Fixed file: own file	Fixed file: new arrangement	Fresh text: own text	Fresh text: new arrangement
Regular, random	0.11	1.25	0.55	0.55
Regular, B 3.6 (fresh: 3.5)	0.10	1.26	0.54	0.54
Regular, grouped	0.09	0.85	0.35	0.35
No stable feature, grouped	0.09	1.00	0.39	0.39
Objects, same-object runs, grouped	0.08	0.66	0.28	0.28
Objects, different-object runs, grouped	0.09	0.76	0.32	0.32

D Data and code

All code, corpora, test items, model scores and analysis outputs, and the final weights and training logs of every trained model, will be released on publication; anonymised copies of the code and results are in the supplementary material. A README gives the command that regenerates each table and figure, with MD5 checksums for the fixed inputs. The pretrained models are public at the revisions listed.

E Language, corpora and earlier measures

This appendix describes the artificial language and the measures of the earlier experiments.

E.1 Corpora

Each corpus describes 32 kinds of object, each named by a novel noun (*wug*, *dax*, *blorp* and so on), with 14 exemplars per kind. Each exemplar has a shape, a colour and a texture, each a novel word from a list of ten per dimension, and is described by one sentence in one of five frames:

1. *A wug is a [shape] [colour] [texture] thing.*
2. *Look at the wug. It is [shape] and [colour].*
3. *Here is another wug. This one is [shape].*
4. *The wug I showed you is [shape], but a different [colour].*

5. *That is a wug because it is [shape].*

Shape is named in all five frames, colour in three and texture in one. The frames are the same in every corpus we compare, so they cannot by themselves explain a difference between corpora, although they may interact with what is manipulated. The 448 sentences of a corpus, each used once, make up one pass. Five corpora were generated for each condition, one per seed.

Content. The corpora differ in which feature, if any, stays fixed within a kind (Table eftab:1). In **Regular** corpora every kind keeps one shape across its 14 exemplars, while colour and texture vary. In **Mixed-slot** corpora each kind keeps one word from the shape list, but for about half the kinds that word appears in the texture slot, and colour varies; no single dimension is fixed across all kinds. In **No stable feature** corpora, built from Regular, each exemplar's shape is drawn at random from the list, so nothing is fixed within a kind. In **Colour stable** corpora, also built from Regular, shape is drawn in the same way and each kind keeps one colour instead. Frame 4 is unchanged, so in Colour stable corpora it names the kind's fixed colour after the words *a different*.

Objects. A further set of corpora separates two kinds of nearby repetition. In the Regular corpora, two sentences about a kind always describe different exemplars. In speech, nearby mentions often return to the same object. The Objects corpora keep the Regular content but give each kind 13 objects and describe each object four times, once in each of frames 1, 2, 3 and 5. Frame 4 is left out because it states that the colour differs. A pass of these corpora has 1,664 sentences.

Order. The same sentences were arranged in several orders. In **random** order every pass is shuffled. In **grouped** order, within each pass, the sentences about each kind are shuffled, cut into runs of four, and the runs are shuffled across kinds. Every sentence occurs exactly as often as in random order. In **graded** orders each run is kept with a set probability and otherwise split into single sentences before shuffling, which gives intermediate levels of clustering. In the Objects corpora, a **same-object run** is the four descriptions of one object, and a **different-object run** is four different objects of one kind, one per frame. A Latin square over objects and frames ensures that every sentence is used once per pass in every order. Same-object and different-object runs were each arranged fully grouped and at an intermediate level.

Reassigned. In every corpus so far, a kind's fixed feature is fixed for the whole corpus, so the noun alone predicts it, and a model can predict the shape in a sentence from its noun without comparing the sentence with the one before. In Reassigned corpora, built from Regular, each kind receives a new shape, drawn at random from the ten, in every pass, and keeps it in all of its sentences in that pass. The next sentence about the same kind still repeats the shape, but the noun no longer says which shape it will be. These corpora were arranged in random and grouped order.

Fresh and fixed text. A model trained many times over on one short file can memorise it. Our main models were therefore trained on fresh text: 7,500 passes of a corpus (2,100 for the Objects corpora), each arranged with its own random draw by the rule of its order, 34 to 35 million tokens in all. Training read this text in non-overlapping 128-token windows, so no window was seen twice. A second set of models was trained in the common way, on one fixed file of 11,648 sentences (26 passes, or 7 for the Objects corpora; about 120,000 tokens), with windows starting at random positions, so that each position was seen 160 to 290 times. These fixed-file models memorised their files. Their loss was 0.08 to 0.11 nats per token on their own file but 0.66 to 1.26 on a new arrangement of the same sentences, while fresh-text models scored 0.28 to 0.55 on both (Table C2). Table 1 of the long version shows which conditions were trained on fresh text and which on a fixed file; where both exist, we report fresh text first. The fresh graded order, built by the same rule as the fixed one at B = 3.6, measured B = 3.5.

Clustering. We measure clustering with one statistic, B, here and in Study 2. Take the sequence of mentions of object nouns in a text, and every pair of consecutive mentions that are no more than 100 words apart. P is the share of these pairs that name the same kind. P_0 is the same share after the sentences (or utterances) of each document are shuffled, averaged over shuffles. B = P/P_0, pooled over documents. For example, in *Look at the dog. The dog is big. Where is the cup? Here is another dog.*, the consecutive pairs are dog–dog, dog–cup and cup–dog, so P = 1/3. B = 1 means no clustering beyond what the content predicts. Random-order Regular corpora have B = 0.9, slightly below 1, because each pass is shuffled

separately: within a pass, a sentence's kind has 13 other members among 447 sentences, against 363 among 11,647 when the whole corpus is shuffled. The graded levels (Table eftab:1) were chosen after the clustering of natural input had been measured (Study 2): 1.4 and 3.6 match B in written text (1.43) and in speech to children (3.57). They were fixed from the corpora alone before any graded-order model was trained.

Table I1: Study 1 corpora. Every corpus describes 32 kinds. Fresh: 7,500 passes (2,100 for Objects), each arranged anew and read once in non-overlapping 128-token windows. Fixed: one file of 11,648 sentences (10,304 to 12,992 for three further controls: No kind label, Paraphrased and Weak label), read in 128-token windows at random positions unless marked. Each order of each corpus was trained at three sizes with five seeds (15 models). The further controls are reported in Appendix D (Table D1).

Corpus	What stays fixed within a kind	Orders (measured B)	Text	Models
Regular	Shape. Colour and texture vary.	Random (0.9); graded (3.5); grouped (23)	Fresh	45
No stable feature	Nothing	Grouped (23)	Fresh	15
Objects	Shape; each object described four times	Same-object runs (24); different-object runs (24)	Fresh	30
Reassigned	Shape within a pass; a new shape for each kind in every pass	Random (0.9); grouped (23)	Fresh	30
Regular	Shape	Random (0.9); graded (1.4, 3.6, 7, 12); grouped (23)	Fixed	90
Regular	Shape	Random (0.9); graded (3.6)	Fixed, 512-token window	30
Mixed-slot	One shape-list word, in the shape slot for about half the kinds and the texture slot for the rest	Random; grouped	Fixed	30
No stable feature	Nothing	Random; grouped; random with a 512-token window	Fixed	45
Colour stable	Colour. Shape varies.	Random; grouped	Fixed	30
Objects	Shape; each object described four times	Random (1.0); same-object runs (3.6, 24); different-object runs (3.6, 24)	Fixed	75
Further controls: no kind label, weak label, paraphrased, frequency matched, added noise, feature swap	As Regular, with one change each	Random	Fixed	90
Total				510

E.2 Models and training

We trained decoder-only transformers of three sizes: 3.4 million parameters (4 layers, 4 heads, width 256), 11.0 million (6 layers, 8 heads, width 384) and 25.6 million (8 layers, 8 heads, width 512). They were trained for 5,000, 8,000 and 10,000 steps with batches of 32, 32 and 16, peak learning rates of 3×10^{-4}, 2×10^{-4} and 1×10^{-4}, AdamW (betas 0.9 and 0.95, weight decay 0.1), linear warm-up, cosine decay and dropout of 0.1. Each training example is a 128-token window, which spans several sentences, so attention crosses sentence boundaries. The tokeniser for the Regular corpora has 351 entries (314 to 339 for the other corpus families), and every artificial word is a single token. A seed sets the corpus, the initial weights and the batch order (the long version lists the models per condition). For the window test, models were trained with 512-token windows and a quarter of the batch size, so that every step saw the same number of tokens. All models have separate input and output embeddings, and the parameter counts include both; the model code ties the two, but the tie was lost on the training hardware (Appendix A). Table C1 lists every setting.

E.3 Measuring the expectation

Eight novel nouns that never occur in training (*snerg*, *twick* and so on) are used in test items. They are in the vocabulary, so each is one token, but their input embeddings were never trained; weight decay shrinks them by less than 10% over training, so they remain distinct random vectors. With the intended tie, the same vectors would have served as the novel nouns' output weights and would have changed in training; we cannot say whether the models would then have tracked these nouns. Novel nouns keep learned knowledge about a noun out of the test, but the test then depends on the model registering that the same untrained token recurs. We therefore also test with nouns from training, whose embeddings are trained (Appendix F). Each item gives the model one sentence in frame 1 describing an exemplar, followed by two continuations that differ in one word:

> Context: *A snerg is a krannish pulek skorli thing.* Repeat: *A snerg is a* ***krannish*** *temil tramel thing.* New: *A snerg is a* ***tova*** *temil tramel thing.*

In this item the probed dimension is shape. The repeat continuation keeps the context's shape, the new one does not, and both give new values for the other two dimensions. The model scores each continuation by the sum of the log probabilities of its tokens given the context, so the difference, Δ, includes the changed word and its effect on the words after it. Δ is the model's preference for repeating the probed dimension, in nats. Trained models generally prefer words they have just seen, so Δ alone mixes the dimension with copying. Our index is the difference between dimensions:

$$\text{index} = \Delta_{\text{shape}} - \Delta_{\text{colour}}$$

Copying that affects shape and colour equally cancels. Copying that differs between dimensions does not, which is one reason every conclusion rests on comparisons between corpora tested in the same way, and why we report Δ for each dimension as well. The shape-over-texture index is defined in the same way. With one exemplar, nothing in the context shows which dimension is fixed, so differences between trained conditions come from training.

Every item is also given with a different novel noun in the continuations (*A yeff is a krannish temil tramel thing*). The **noun-specific index** is the index after the same noun minus the index after a different noun. A kind-tied expectation scores above zero on it; a noun-blind one scores zero.

With 25 draws of features per noun, three probed dimensions and two continuation nouns there are 1,200 one-exemplar items. A further 2,400 items with two or four exemplars test the use of evidence in the context (Table E2). Untrained models of the same architecture, with the Regular tokeniser, give an index of 0.02 nats (SD 0.07).

E.4 Statistics

Each model gives one index, its mean over items. Models of the three sizes trained with the same seed share a corpus, so the five seeds are the units of replication. We average the three sizes within each seed and compare conditions by paired t tests over the five seed means (4 degrees of freedom); the differences and 95% intervals given in the text are these seed-level estimates. Registered contrasts were tested one-sided, with Holm (1979) correction within each registered family. For fresh-text models the registered test was this seed-level test. For fixed-file models it was a linear mixed model (index $\sim$ condition + size, with a random intercept for seed) or, for some contrasts, a Welch t test over models; Table E1 gives the seed-level results beside them. Single conditions were compared with zero by one-sample t tests. Results reported with a Holm-corrected p, or called registered, were registered; all others are exploratory. We registered 0.2 nats as the smallest effect of interest and tested equivalence to zero against it with two one-sided tests (Lakens, 2017). This bound is small next to the effects reported here, so the equivalence tests are strict.

F Test-noun representation and changed-word scoring

These analyses were added after the registrations and are exploratory. They rescore the 120 fresh-text models on the one-exemplar items (`scripts/recut_rescore.py`,

`scripts/recut_rescore_analysis.py`). **Typical tokens**: each novel noun's input embedding is replaced by the mean input embedding of the 32 training nouns plus $r\,u_j$, where r is the mean distance of the training-noun embeddings from that mean and the u_j are orthonormal random directions (fixed seed), so the eight tokens look like ordinary nouns but stay distinct; output weights are unchanged. **Changed-word** Δ: the difference in log probability of the one word that differs between the continuations, given the identical prefix. With the original embeddings and whole-continuation Δ, the rescoring reproduced the reported scores (144,000 item scores; largest difference 2×10^{-5} nats). Trained-noun values are from the registered trained-noun items (long version in the supplementary material; `scripts/recut_noun_store_check.py`).

Table L1: Index and noun-specific index under four scorings (nats; mean of seed means). Original: unseen tokens with untrained embeddings, as reported; typical: the same tokens with typical embeddings; full: whole-continuation Δ, as reported; word: changed word only.

	Index				Noun-specific index			
Condition	orig./full	typ./full	orig./word	typ./word	orig./full	typ./full	orig./word	typ./word
Regular, random	−1.56	−1.54	−1.26	−1.50	−0.05	−0.31	0.01	−0.20
Regular, graded (3.5)	1.28	−1.41	1.45	−1.25	−0.03	−0.15	0.03	−0.04
Regular, grouped	9.52	1.46	9.65	1.38	0.12	0.00	0.12	0.01
No stable feature, grouped	−3.98	−3.51	−3.74	−3.24	−0.35	−0.26	−0.34	−0.21
Objects, same-object runs	5.87	−1.85	6.06	−1.87	−0.07	0.14	−0.06	0.06
Objects, different-object runs	7.60	−1.25	7.59	−1.33	−0.01	0.19	−0.01	0.11
Reassigned, random	0.21	0.18	0.32	0.27	0.02	−0.10	0.03	−0.01
Reassigned, grouped	9.80	3.87	9.94	3.98	0.37	0.70	0.37	0.74

Table L2: Preference for repeating the context's shape and colour after one exemplar, same noun, for the three kinds of test noun (nats; whole-continuation Δ).

	Δ shape			Δ colour		
Condition	Unseen	Typical	Trained	Unseen	Typical	Trained
Regular, random	−1.55	−1.52	0.03	0.01	0.02	0.05
Regular, graded (3.5)	1.18	−1.52	0.06	−0.10	−0.12	0.19
Regular, grouped	6.88	−0.72	0.06	−2.64	−2.18	−1.94
No stable feature, grouped	−3.88	−3.51	−3.58	0.10	0.00	0.17
Objects, same-object runs	6.87	−1.17	−0.09	0.99	0.68	0.46
Objects, different-object runs	8.05	−1.14	0.04	0.46	0.11	0.01
Reassigned, random	0.22	0.24	0.24	0.02	0.06	0.10
Reassigned, grouped	6.81	1.17	10.01	−2.99	−2.70	−1.65

Table L3: Noun-familiarity curve: unseen tokens' input embeddings blended from their untrained values (0) to typical embeddings (1). Index and Δ shape, same noun (nats; mean of seed means); slope per unit blend with 95% interval over five seeds.

Condition	Measure	0	0.25	0.5	0.75	1	Slope
Regular, random	Index	−1.56	−1.83	−1.93	−1.79	−1.54	0.03 [−0.22, 0.28]
Regular, random	Δ shape	−1.55	−1.83	−1.92	−1.78	−1.52	0.05 [−0.26, 0.35]
Regular, grouped	Index	9.52	6.70	4.32	2.10	1.46	−8.28 [−9.87, −6.70]
Regular, grouped	Δ shape	6.88	4.27	2.08	−0.08	−0.72	−7.82 [−9.59, −6.04]
Objects, same-object runs	Index	5.87	3.71	1.08	−1.41	−1.85	−8.23 [−10.05, −6.40]
Objects, same-object runs	Δ shape	6.87	4.59	1.90	−0.65	−1.17	−8.52 [−10.22, −6.83]
Objects, different-object runs	Index	7.60	4.53	1.87	−0.82	−1.25	−9.22 [−9.97, −8.46]
Objects, different-object runs	Δ shape	8.05	4.85	2.08	−0.69	−1.14	−9.57 [−10.14, −9.00]
Reassigned, grouped	Index	9.80	8.03	5.34	3.87	3.87	−6.41 [−8.74, −4.08]
Reassigned, grouped	Δ shape	6.81	4.83	2.24	0.97	1.17	−6.06 [−7.28, −4.84]

Mean input-embedding norms: trained nouns 0.55, unseen tokens 0.36, typical tokens 0.55.

G The familiarity study (Amendment 9)

Registration. The design, hypotheses and analysis were registered before any corpus was built (Amendment 9). An addendum, written the same day before any corpus was built, fixed details the registration left open (36 context draws per noun; uniform frame proportions; exposures placed at evenly spaced read positions from 5% to 95% of training, after warm-up for every size; random in-window position; the 40 nonce nouns; how the pilot fixes the number of nouns per cell; how a hypothesis counts as supported; the gate for H35; and that heads without prefix matching are described as copying heads). A pilot (one seed, 3.4M models, never used in a test) set three nouns per cell; with the registered rule the expected power for a 1-nat difference was 0.57, below the planned 0.8, which we report as a shortfall. Corpus checksums were recorded before training and matched on the training machine; a checker verified every exposure, window and frame from the tokens. All models were trained on one machine.

Design. In every model, 40 nonce nouns fill 13 cells (three per cell, four unseen): fixed shape, uninformative shape (a new shape in every sentence, drawn from the corpus's shape distribution) and shape-absent (a sixth frame, *The noun is colour and texture*, also used by trained kinds in one sentence in fifteen), each at 10, 100, 1,000 and 10,000 exposures, plus unseen. Nouns rotate over cells across ten seeds. Each familiarity sentence lies wholly inside one 128-token window, at most one per window, between runs in grouped corpora. Base corpora: Regular grouped, Regular random, Reassigned grouped; sizes 3.4M, 11.0M and 25.6M; ten seeds; 90 models with untied embeddings. Measures are read from the full log-softmax over the ten shape words at the position that predicts the shape in *A noun is a*: carry $= L(s) - \bar{L}$ after a context sentence with shape s; stored preference $= L(s^*) - \bar{L}$; knowledge $= L(s^*)$ minus the mean over the other nine shapes after an unrelated sentence, where $\bar{L}$ is the mean over shapes other than s and s^*.

Table M1: Registered tests of the familiarity study (Regular grouped models unless stated; seed-level estimates over ten seeds, sizes averaged within seed; nats, proportions for H33a, attention probability for H35). Holm p is adjusted over H30 to H34; H35 was not entered because H33(a) failed. Sizes: number of the three model sizes with the registered direction.

	Test	Estimate	95% CI	Holm p	Sizes	Verdict
M1	Knowledge at 1,000 exposures, fixed shape (each corpus)	9.53; 8.61; 8.26				Passed
H30	Carry falls faster with exposure for fixed-shape than shape-absent nouns (slope difference)	0.16	[0.01, 0.30]	1.000	0	Not supported
H31	Shape-absent (1k and 10k) minus unseen with typical embedding (two-sided; TOST ± 1)	0.33	[0.13, 0.52]	.013	3	Different, and equivalent within 1 nat
H32	Stored preference rises with exposure, fixed shape (slope)	2.97	[2.79, 3.16]	$<.001$	3	Supported
H33a	Ablating candidates removes more than half of unseen carry (proportion)	0.22	[0.07, 0.36]	1.000	2	Not supported
H33b	Random heads minus candidates, unseen carry after ablation	0.64	[−0.43, 1.71]	1.000	2	Not supported
H34	Reassigned: trained-kind carry intact minus candidates ablated	7.41	[6.02, 8.80]	$<.001$	3	Supported
H35	Attention slope difference, fixed shape minus shape-absent	0.002	[0.000, 0.005]	–	2	Not tested (gate)
R1	Replication: grouped minus random Regular, unseen carry	4.68	[3.71, 5.65]	$<.001^a$	3	Replicated

[a] Outside the Holm family (replication); unadjusted one-sided p. M1 values: Regular grouped; Regular random; Reassigned grouped (all with slopes above zero, $p < .001$). H31 TOST $p < .001$.

Table M2: Carry (nats) by noun type and exposure, mean of seed values (sizes averaged). Unseen: never in training; typical: the unseen nouns with a typical embedding; trained: the 32 base kinds.

Corpus	Unseen	Typical	Trained	Familiar nouns	10	100	1k	10k
Regular grouped	4.53	−0.33	−0.02	Fixed shape	3.70	0.30	−0.30	−0.37
				Uninformative	3.57	0.16	0.00	−0.01
				Shape-absent	4.47	0.86	0.05	−0.05
Reassigned grouped	6.85	1.57	9.98	Fixed shape	6.40	2.13	0.70	0.60
				Uninformative	6.34	2.53	0.30	0.15
				Shape-absent	6.97	4.89	2.04	1.86
Regular random	−0.15	−0.72	−0.10	Fixed shape	−0.13	−0.13	−0.06	−0.05
				Uninformative	−0.10	−0.03	−0.02	−0.01
				Shape-absent	−0.19	−0.09	−0.07	−0.05

Table M3: Knowledge and stored preference (nats) in Regular grouped models, mean of seed values.

Type	Measure	10	100	1k	10k
Fixed shape	Knowledge	1.59	8.02	9.53	10.61
	Stored	1.41	7.74	9.56	10.71
Uninformative	Knowledge	0.04	−0.03	0.00	0.03
	Stored	0.01	−0.03	−0.01	0.03
Shape-absent	Knowledge	−0.14	−0.17	0.12	0.08
	Stored	0.11	−0.16	0.10	0.06

Table M4: Head analysis by corpus and size (means over ten seeds). Candidates were the k heads with the highest product of prefix-matching and copying scores; in grouped models they were in layer 0 in all but one model (a 3.4M Regular grouped model chose one layer-1 head), and in random-order models a few were in later layers. In the 3.4M and 11.0M models layer 0 has exactly k non-candidate heads, so the random draws repeat one set. Prefix: highest prefix-matching score of any head (uniform attention gives about 0.027). Carry of unseen nouns and of trained kinds intact, with the candidates mean-ablated, and with random heads ablated (mean of 200 or 20 draws). DLA: change in unseen carry predicted by the candidates' direct paths.

				Unseen carry			Trained carry			
Corpus	Size	k	Prefix	Intact	Cand.	Random	Intact	Cand.	Random	DLA
Regular grouped	3.4M	2	0.028	8.10	−0.42	5.26	−0.25	−0.85	−0.71	−0.04
	11.0M	4	0.037	2.98	5.80	1.95	0.08	0.17	−0.40	−0.01
	25.6M	6	0.037	2.50	0.73	0.81	0.12	−0.60	−0.18	−0.01
Reassigned grouped	3.4M	2	0.029	6.48	4.22	3.27	9.59	2.83	3.37	−0.12
	11.0M	4	0.032	7.21	6.56	4.82	10.10	5.02	2.45	−0.03
	25.6M	6	0.031	6.88	0.26	3.26	10.26	−0.11	3.39	−0.01
Regular random	3.4M	2	0.028	−0.24	−0.09	−0.34	−0.08	0.00	−0.11	−0.01
	11.0M	4	0.030	0.04	0.25	0.03	−0.06	0.04	−0.11	−0.01
	25.6M	6	0.030	−0.25	0.00	−0.37	−0.18	−0.10	−0.27	−0.01

Table M5: Attention of the candidate heads from the position that predicts the shape to the context's shape word (mean over candidates; mean of seed values; exploratory).

Corpus	Type	Unseen	10	100	1k	10k
Regular grouped	Fixed shape	0.155	0.144	0.082	0.071	0.071
	Uninformative		0.147	0.088	0.076	0.074
	Shape-absent		0.154	0.097	0.077	0.075
Reassigned grouped	Fixed shape	0.219	0.217	0.108	0.055	0.059
	Uninformative		0.217	0.157	0.071	0.070
	Shape-absent		0.219	0.170	0.115	0.139
Regular random	Fixed shape	0.089	0.088	0.076	0.067	0.067
	Uninformative		0.087	0.082	0.075	0.073
	Shape-absent		0.089	0.084	0.075	0.073

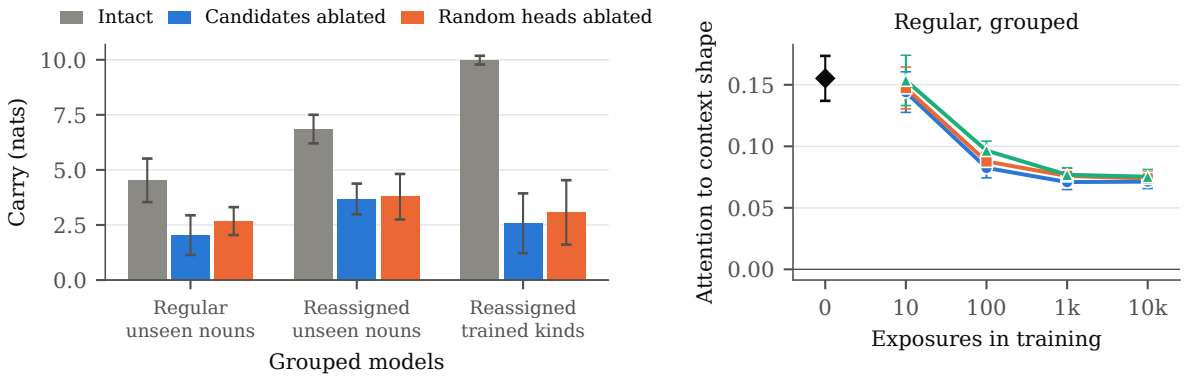


Figure M1: Experiment 1. Left: carry intact, with the candidate heads mean-ablated, and with random heads ablated (nats; mean over ten seeds and three sizes, 95% intervals over seeds). Right: the candidates' attention from the position that predicts the shape to the context's shape word, grouped Regular models (exploratory).

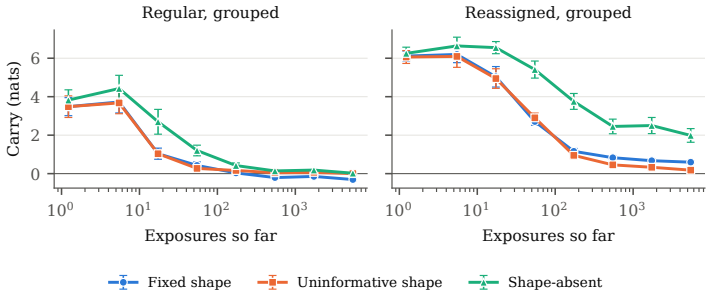


Figure M2: Experiment 1. Carry against the exposures each noun had received by each saved checkpoint (exploratory; nats; mean over ten seeds of size-averaged values, 95% intervals). Exposure is confounded with training time.

H Recurrence, matched frames, embedding rows and pretrained models (Amendment 10)

Registration. Amendment 10 has three parts: Part A is Experiment 2, Part B Experiment 3 (pretrained models) and Part C the row analysis. It was registered after the Amendment 9 results, after its corpora were built and checksummed, and before any of them was used, any of its models was trained, or any pretrained model was run on its items. It answered two confounds raised by a blind review of the Amendment 9 results (by an AI assistant given only the write-up of Experiment 1): familiarity nouns never recurred within a window, and shape-absent nouns never appeared in a frame with a shape slot after *is a*. The noun lists for Part B were fixed, with checksums, before the pilot. A pilot on Pythia-410M (never in a test) gave a mean increase of 0.13 nats from the original to the atypical_init row, so the smallest effect of interest for H48 was 0.30 nats by the registered rule. One deviation: some Pythia tokenizers define no padding token, so the scoring script falls back to token id 0; padded positions are masked and only the last real token is read, and batched and single-item scores agreed exactly. No score had been produced by the affected models.

Part A design. Everything not stated is as in Amendment 9 (Appendix G). Histories: *fixed*, frame 1 (*A noun is a shape colour texture thing*) with the noun's own shape; *absent*, frame 7 (*A noun is a colour texture thing*), which names no shape; *rotating*, frame 1 with a shape drawn from the corpus's shape distribution, a new one for every sentence when presented singly and one per run, changing between runs, when presented in runs. Presentation: *single*, one sentence per unit; *run*, four consecutive sentences about the noun in one window. Exposures (100 or 1,000) count sentences, so runs and singles are matched in sentences, not in units or spacing. Trained kinds use frame 7 in one sentence in fifteen, so the frame is not new. Three nouns per cell (12 cells) and four unseen nouns; Regular grouped (confirmatory) and

Reassigned grouped (exploratory) base corpora; three sizes and ten seeds (60 models). A checker re-read every corpus and verified exposures, units, frames, shapes within runs and window placement; all 40 corpora matched checksums recorded before training.

Table N1: Registered tests of Part A (Regular grouped models; 100 and 1,000 exposures pooled; seed-level estimates over ten seeds, sizes averaged within seed; nats). Holm p is adjusted over H40 to H44. Sizes: number of the three sizes with the estimate's direction. Readings follow the registered table: "above" means the Holm threshold was met and the direction held at two of three sizes; TOST p for H40, H41 and H44 was above .99, so none was equivalent.

	Test	Estimate	95% CI	Holm p	Sizes	Reading
M2	Unseen carry minus 1 nat (manipulation check)	5.35	[4.59, 6.12]	$<.001^a$		Passed
K1	Knowledge, fixed runs minus fixed singles (equivalence ± 1)	−1.09	[−1.45, −0.73]	$<.001^a$		Not equivalent (H41 not interpreted)
H40	Absent runs minus absent singles (two-sided; TOST ± 0.3)	0.80	[0.47, 1.13]	$<.001$	3	Above
H41	Fixed runs minus fixed singles (two-sided; TOST ± 0.3)	1.69	[1.29, 2.08]	$<.001$	3	Above (not interpreted)
H42	Rotating runs minus rotating singles (one-sided)	3.39	[2.69, 4.09]	$<.001$	3	Supported
H43	Fixed singles minus absent singles (one-sided)	−1.02	[−1.27, −0.76]	$<.001$	3	Supported
H44	Absent singles minus unseen with typical row (two-sided; TOST ± 0.3)	1.42	[1.12, 1.73]	$<.001$	3	Above

[a] Outside the Holm family; unadjusted p. The M2 estimate is unseen carry minus 1 nat, so unseen carry was 6.35 nats.

Table N2: Carry (nats) by cell, mean of seed values (sizes averaged) with 95% intervals over ten seeds. Unseen nouns have no exposures; the typical row replaces an unseen noun's input row with one typical of trained nouns.

Base	Noun	100 exposures	1,000 exposures
Regular, grouped	Fixed shape, singles	1.52 [1.18, 1.86]	0.52 [0.36, 0.69]
	Fixed shape, runs	4.43 [3.49, 5.36]	0.99 [0.68, 1.30]
	Shape-absent, singles	3.09 [2.56, 3.63]	0.98 [0.76, 1.20]
	Shape-absent, runs	4.65 [3.94, 5.36]	1.03 [0.59, 1.47]
	Rotating shape, singles	1.59 [1.30, 1.88]	0.50 [0.38, 0.62]
	Rotating shape, runs	5.02 [4.18, 5.86]	3.85 [2.93, 4.77]
	Unseen	6.35 [5.59, 7.12]	
	Unseen, typical row	0.61 [0.28, 0.95]	
Reassigned, grouped	Fixed shape, singles	2.23 [1.92, 2.55]	0.44 [0.37, 0.51]
	Fixed shape, runs	5.35 [4.88, 5.82]	3.57 [3.02, 4.11]
	Shape-absent, singles	1.83 [1.49, 2.17]	0.63 [0.53, 0.72]
	Shape-absent, runs	3.79 [3.38, 4.20]	2.09 [1.82, 2.35]
	Rotating shape, singles	2.37 [2.04, 2.71]	0.51 [0.38, 0.64]
	Rotating shape, runs	5.40 [4.95, 5.86]	6.82 [6.48, 7.17]
	Unseen	5.02 [4.61, 5.42]	
	Unseen, typical row	0.43 [0.32, 0.55]	

Table N3: Knowledge of the noun's own shape and expectation of a shape rather than a colour at the query position (slot: log of the summed probability of the ten shapes minus that of the ten colours), Regular grouped models, mean of seed values (exploratory). Unseen nouns: knowledge −0.04, slot 1.49.

	Knowledge		Slot	
Noun	100	1,000	100	1,000
Fixed shape, singles	5.37	7.80	4.03	6.16
Fixed shape, runs	3.02	7.98	3.28	6.24
Shape-absent, singles	0.00	−0.09	−2.98	−4.93
Shape-absent, runs	0.05	−0.25	−1.12	−4.43
Rotating shape, singles	−0.08	0.01	5.04	5.30
Rotating shape, runs	0.14	0.06	3.74	5.24

Table N4: The Part A contrasts in Reassigned grouped models (exploratory; seed-level, unadjusted two-sided p).

	Contrast	Estimate	95% CI	p
H40	Absent runs minus absent singles	1.71	[1.36, 2.06]	<.001
H41	Fixed runs minus fixed singles	3.12	[2.71, 3.53]	<.001
H42	Rotating runs minus rotating singles	4.67	[4.28, 5.06]	<.001
H43	Fixed singles minus absent singles	0.11	[−0.07, 0.29]	.196

Table N5: Matched contrasts in Part A (exploratory; seed-level, nats, 95% intervals). Rotating minus fixed runs: same frame, presentation and number of occasions; the two differ only in whether the shape changes between runs (and so in whether a shape is stored). Fixed minus rotating singles: same frame, presentation and expectation of a shape at the test position (Table N3); they differ only in whether the noun has one stored shape.

Base	Contrast	Pooled	100	1,000
Regular grouped	Rotating runs − fixed runs	1.73 [1.25, 2.20]	0.59 [0.23, 0.95]	2.86 [1.99, 3.74]
	Fixed singles − rotating singles	−0.02 [−0.16, 0.11]	−0.07 [−0.29, 0.15]	0.02 [−0.04, 0.08]
	Fixed runs − fixed singles	1.69 [1.29, 2.08]	2.91 [2.21, 3.60]	0.47 [0.27, 0.66]
Reassigned grouped	Rotating runs − fixed runs	1.65 [1.35, 1.96]	0.05 [−0.05, 0.16]	3.26 [2.64, 3.87]
	Fixed singles − rotating singles	−0.10 [−0.23, 0.02]	−0.14 [−0.34, 0.06]	−0.07 [−0.20, 0.05]
	Fixed runs − fixed singles	3.12 [2.71, 3.53]	3.12 [2.61, 3.62]	3.13 [2.59, 3.66]

Knowledge, fixed runs minus fixed singles (Regular grouped): −2.35 [−2.82, −1.88] at 100 exposures, 0.18 [−0.18, 0.54] at 1,000.

Table N6: Noun-specific carry (exploratory): carry when the context sentence is about the query noun minus carry when it is about another noun of the same cell, with the same shape (nats; seed-level means). Near zero means the model copies the context's shape whichever noun the context was about.

		Regular grouped		Reassigned grouped	
Experiment	Noun	100	1,000	100	1,000
2	Unseen	0.05		0.48	
	Fixed, singles	0.01	−0.02	0.22	0.06
	Fixed, runs	0.06	−0.05	0.58	0.65
	Shape-absent, singles	0.07	0.05	0.25	0.05
	Shape-absent, runs	0.01	0.00	0.46	0.70
	Rotating, singles	0.04	0.01	0.35	0.05
	Rotating, runs	0.05	0.05	0.58	1.45
1	Unseen	0.14		0.45	
	Fixed shape	−0.05	−0.02	0.21	0.01
	Uninformative	0.00	0.00	0.51	0.04
	Shape-absent	0.05	0.00	0.45	0.22

Table N7: Part A by model size (Regular grouped; seed means, 100 and 1,000 exposures pooled; nats).

Size	Unseen	Typical row	H40	H41	H42	H43	H44	Rot. − fixed runs
3.4M	7.67	−0.30	0.65	2.41	5.05	−0.84	1.67	2.94
11.0M	5.74	0.64	1.23	1.46	2.82	−0.78	1.25	1.24
25.6M	5.65	1.51	0.53	1.20	2.29	−1.43	1.36	1.00

Part C (embedding rows in the Amendment 9 models). In the 60 final grouped models of Amendment 9 we replaced the input rows of unseen nouns and scored carry again; the unmodified condition reproduced the Amendment 9 scores. Conditions: own row; own direction scaled to the mean norm of the trained kinds' rows (ownnorm); a random direction at that norm (randnorm); the trained kinds' mean row. Shape-absent nouns at 10,000 exposures were also scored with their own direction scaled to the unseen nouns' mean norm (shrunk).

Part B (pretrained models). 44 nouns in 22 pairs, matched on Pile frequency and count/mass type, were coded for colour typicality by two independent AI passes with a fixed prompt before any model was run (exact agreement 94%, weighted $\kappa = 0.96$); nouns that are also colour terms or that both passes flagged as polysemous were excluded. Colour-typical nouns are mostly natural kinds, so kind was a

Table N8: Part C: carry (nats) by input row, mean over seeds (sizes averaged). H49 (registered, Regular grouped): ownnorm minus own −0.52 [−0.85, −0.20], one-sided p = .003, direction at 2 of 3 sizes, supported. In Reassigned models the same contrast went the other way (0.27 [0.12, 0.41]; exploratory). Randnorm minus ownnorm: 0.09 (Regular), 0.03 (Reassigned).

	Unseen nouns				Shape-absent, 10k	
Corpus	Own	Ownnorm	Randnorm	Trained mean	Own	Shrunk
Regular grouped	4.53	4.00	4.09	−0.44	−0.05	0.53
Reassigned grouped	6.85	7.12	7.15	1.05	1.86	4.16

covariate. Each noun was read in three neutral frames with each of the eleven basic colour terms that are one token; carry is the mean over frames and ordered colour pairs of $\log p(c_1 \mid c_1) - \log p(c_1 \mid c_2)$, with colour probabilities renormalised over the colour set. Only the noun's input row was changed.

Table N9: Part B registered tests by primary model (nouns or pairs as units; Holm within model over H45 to H48). H45: atypical_init minus original; H46: atypical_init minus typical row; H47: colour-typical minus colour-variable nouns, original rows; H48: difference between groups in the H45 increase (TOST against ±0.30). A hypothesis needed Holm in at least four of the five models. Outcome: H45 and H46 met Holm in two of five models, H47 in none, and H48 was inconclusive (equivalent in one model).

	H45		H46		H47		H48	
Model	Est.	Holm	Est.	Holm	Est.	Holm	Est.	TOST p
1B	0.20	met	0.21	met	0.00	–	0.07	.097
1.4B	0.34	met	−0.22	–	−0.01	–	−0.08	.021
2.8B	−0.38	–	0.08	–	−0.16	–	0.11	.222
6.9B	0.07	–	0.36	met	−0.09	–	0.10	.127
12B	−0.35	–	−0.21	–	0.11	–	0.08	.066

Mixed model over the five primary models (carry ∼ condition × group + model + kind; random intercepts for noun and pair, random condition slopes by noun): atypical_init 0.00 (p = .956), typical row −0.04 (p = .559), colour-variable group 0.06 (p = .527).

Table N10: Part B: carry of the context's colour (nats; mean over nouns) by model and input row, and with original rows by group. Atypical: the atypical_init direction at the noun's own norm (exploratory). 70M and 160M are descriptive. Prior entropy: entropy of the colour distribution after *The noun was* with no context (nats; uniform over eleven colours is 2.40).

Model	Original	Atypical_init	Atypical	Typical row	Colour-typical	Colour-variable	Prior entropy
70M	1.59	1.40	1.41	1.16	1.61	1.56	2.09
160M	0.93	1.95	1.95	1.03	0.87	1.00	2.01
1B	1.89	2.09	2.09	1.88	1.89	1.89	1.95
1.4B	2.10	2.44	2.44	2.66	2.10	2.10	1.97
2.8B	2.70	2.31	2.31	2.23	2.61	2.78	1.92
6.9B	2.67	2.75	2.75	2.39	2.63	2.72	1.94
12B	2.68	2.33	2.33	2.54	2.73	2.62	1.91

Table N11: Part B checks (exploratory). Stored colour: with no context, log probability of a colour-typical noun's coded colour minus the mean over the other colours after *The noun was* (nats; mean over 22 nouns, and the share of nouns above zero). TOST: equivalence of the H45 and H46 differences within ± 0.30 nats over nouns. Norm only: atypical minus atypical_init (same direction, different norm).

Model	Stored colour	Share > 0	H45 TOST p	H46 TOST p	Norm only
70M	0.72	0.73	.059	.129	0.014
160M	0.90	0.77	1.000	1.000	0.000
1B	1.28	0.82	.125	.091	0.001
1.4B	1.30	0.91	.736	.172	0.000
2.8B	1.55	0.86	.776	.001	0.000
6.9B	1.66	0.95	.012	.801	−0.002
12B	1.73	0.86	.740	.023	−0.001

Table N12: Part B, Pythia-1.4B deduplicated across training (exploratory): carry (nats; mean over nouns) by training step and input row.

Row	1,000	2,000	4,000	8,000	16,000	32,000	64,000	143,000
Original	0.73	1.09	0.72	1.97	1.46	1.67	2.39	2.10
Atypical_init	0.81	1.45	1.10	2.08	1.82	2.11	2.56	2.44
Typical row	0.72	1.06	0.60	1.64	1.17	1.70	2.70	2.66